

Part 1: Sequential Search

Question 1: Successful Search

main.c (C17)

```
#include <stdio.h>
int main() {
    int arr[] = {12, 25, 8, 45, 32, 19, 50};
    int n = 7, key = 32, comps = 0, pos = -1;
    for(int i = 0; i < n; i++) {
        comps++;
        if(arr[i] == key) { pos = i + 1; break; }
    }
    if(pos != -1) printf("Element found at position %d\nNumber of comparisons = %d\n", pos, comps);
    return 0;
}
```

Output terminal

Input Array: 12, 25, 8, 45, 32, 19, 50

Key: 32

Output:

Element found at position 5

Number of comparisons = 5

...Program finished with exit code 0

Question 2: Unsuccessful Search

main.c (C17)

```
#include <stdio.h>
int main() {
    int arr[] = {5, 10, 15, 20, 25, 30, 35};
    int n = 7, key = 18, comps = 0, found = 0;
    for(int i = 0; i < n; i++) {
        comps++;
        if(arr[i] == key) { found = 1; break; }
    }
    if(!found) printf("Element not found\nNumber of comparisons = %d\n", comps);
    return 0;
}
```

Output terminal

Input Array: 5, 10, 15, 20, 25, 30, 35

Key: 18

Output:

Element not found

Number of comparisons = 7

...Program finished with exit code 0

Question 3: First Occurrence

main.c (C17)

```
#include <stdio.h>
int main() {
    int arr[] = {10, 25, 15, 25, 30, 25, 40};
    int n = 7, key = 25;
    for(int i = 0; i < n; i++) {
        if(arr[i] == key) {
            printf("First occurrence at position %d\n", i + 1);
            break;
        }
    }
    return 0;
}
```

Output terminal

```
Input Array: 10, 25, 15, 25, 30, 25, 40
Key: 25
Output:
First occurrence at position 2

...Program finished with exit code 0
```

Question 4: All Occurrences

main.c (C17)

```
#include <stdio.h>
int main() {
    int arr[] = {7, 12, 7, 25, 18, 7, 30, 7};
    int n = 8, key = 7, count = 0;
    printf("Occurrences at positions:\n");
    for(int i = 0; i < n; i++) {
        if(arr[i] == key) {
            printf("%d ", i + 1);
            count++;
        }
    }
    printf("\nTotal occurrences = %d\n", count);
    return 0;
}
```

Output terminal

```
Input Array: 7, 12, 7, 25, 18, 7, 30, 7
Key: 7
Output:
Occurrences at positions:
1 3 6 8
Total occurrences = 4

...Program finished with exit code 0
```

Question 5: Total comparisons, matches, mismatches

main.c (C17)

```
#include <stdio.h>
int main() {
    int arr[] = {3, 6, 9, 12, 15, 18, 21};
    int n = 7, key = 15;
    int comps = 0, matches = 0, mismatches = 0;
    for(int i = 0; i < n; i++) {
        comps++;
        if(arr[i] == key) matches++;
        else mismatches++;
    }
    printf("Total comparisons = %d\nTotal matches = %d\nTotal mismatches = %d\n",
    comps, matches, mismatches);
    return 0;
}
```

Output terminal

```
Input Array: 3, 6, 9, 12, 15, 18, 21
Key: 15
Total comparisons = 7
Total matches = 1
Total mismatches = 6

...Program finished with exit code 0
```

Question 6: Sentinel Sequential Search

main.c (C17)

```
#include <stdio.h>
int main() {
    int arr[8] = {14, 9, 22, 35, 18, 41, 27}; // extra space for sentinel
    int n = 7, key = 18, i = 0, comps = 0;
    arr[n] = key; // set sentinel
    while(arr[i] != key) {
        comps++;
        i++;
    }
    comps++; // for the matching comparison
    if(i < n) printf("Position found = %d\nComparison count = %d\n", i + 1, comps);
    return 0;
}
```

Output terminal

```
Input Array: 14, 9, 22, 35, 18, 41, 27
Key: 18
Output:
Position found = 5
Comparison count = 5

...Program finished with exit code 0
```

Question 7: Student Register Numbers

main.c (C17)

```
#include <stdio.h>
int main() {
    int reg[] = {101, 102, 103, 104, 105, 106};
    int n = 6, key = 104;
    for(int i = 0; i < n; i++) {
        if(reg[i] == key) {
            printf("Register Number found at position %d\n", i + 1);
            break;
        }
    }
    return 0;
}
```

Output terminal

```
Register Numbers: 101, 102, 103, 104, 105, 106
Search Key: 104
Output:
Register Number found at position 4

...Program finished with exit code 0
```

Question 8: String Data Search

main.c (C17)

```
#include <stdio.h>
#include <string.h>
int main() {
    char names[][10] = {"Anu", "Bala", "Charan", "Deepa", "Esha", "Farhan"};
    char key[] = "Deepa";
    for(int i = 0; i < 6; i++) {
        if(strcmp(names[i], key) == 0) {
            printf("Name found at position %d\n", i + 1);
            break;
        }
    }
    return 0;
}
```

Output terminal

```
Names: Anu, Bala, Charan, Deepa, Esha, Farhan
Search Name: Deepa
Output:
Name found at position 4

...Program finished with exit code 0
```

Question 9: 2-D Matrix Search

main.c (C17)

```
#include <stdio.h>
int main() {
    int mat[3][3] = {{12, 8, 15}, {5, 18, 27}, {9, 11, 24}};
    int key = 24, found = 0;
    for(int i = 0; i < 3; i++) {
        for(int j = 0; j < 3; j++) {
            if(mat[i][j] == key) {
                printf("Element found at Row %d Column %d\n", i + 1, j + 1);
                found = 1; break;
            }
        }
        if(found) break;
    }
    return 0;
}
```

Output terminal

```
Matrix:
12 8 15
 5 18 27
 9 11 24
Key = 24
Output:
Element found at Row 3 Column 3

...Program finished with exit code 0
```

Question 10: Performance Analysis

main.c (C17)

```
#include <stdio.h>
int main() {
    printf("Analysis:\n");
    printf("Best-case complexity: O(1)\n");
    printf("Average-case complexity: O(n)\n");
    printf("Worst-case complexity: O(n)\n");
    printf("Space complexity: O(1)\n");
    return 0;
}
```

Output terminal

```
Analysis:
Best-case complexity: O(1)
Average-case complexity: O(n)
Worst-case complexity: O(n)
Space complexity: O(1)

...Program finished with exit code 0
```

Part 2: Brute Force String Matching

Question 1: Pattern Search

main.c (C17)

```
#include <stdio.h>
#include <string.h>
int main() {
    char txt[] = "AABAACAADAABAABA", pat[] = "AABA";
    int n = strlen(txt), m = strlen(pat), comps = 0;
    for(int i = 0; i <= n - m; i++) {
        int j;
        for(j = 0; j < m; j++) {
            comps++;
            if(txt[i+j] != pat[j]) break;
        }
        if(j == m) printf("Pattern found at position %d\n", i + 1);
    }
    printf("Total number of comparisons: %d\n", comps);
    return 0;
}
```

Output terminal

```
Text = "AABAACAADAABAABA"
Pattern = "AABA"
Output:
Pattern found at position 1
Pattern found at position 10
Pattern found at position 13
Total number of comparisons: 31

...Program finished with exit code 0
```

Question 2: All Occurrences

main.c (C17)

```
#include <stdio.h>
#include <string.h>
int main() {
    char txt[] = "BANANABANANA", pat[] = "ANA";
    int n = strlen(txt), m = strlen(pat);
    printf("Occurrences at positions: ");
    for(int i = 0; i <= n - m; i++) {
        int j;
        for(j = 0; j < m; j++) {
            if(txt[i+j] != pat[j]) break;
        }
        if(j == m) printf("%d ", i + 1);
    }
    printf("\n");
    return 0;
}
```

Output terminal

```
Text = "BANANABANANA"
Pattern = "ANA"
Output:
Occurrences at positions: 1 3 7 9
```

...Program finished with exit code 0

Question 3: Comparison Table

main.c (C17)

```
#include <stdio.h>
#include <string.h>
int main() {
    char txt[] = "MISSISSIPPI", pat[] = "ISSI";
    int n = strlen(txt), m = strlen(pat);
    for(int i = 0; i <= n - m; i++) {
        int j, comps = 0;
        for(j = 0; j < m; j++) {
            comps++;
            if(txt[i+j] != pat[j]) break;
        }
        if(j == m) printf("Shift %d | Comps: %d | Match!\n", i, comps);
        else printf("Shift %d | Comps: %d | Mismatch (%c!=%c)\n", i, comps,
txt[i+j], pat[j]);
    }
    return 0;
}
```

Output terminal

```
Text = "MISSISSIPPI"
Pattern = "ISSI"
Shift 0 | Comps: 1 | Mismatch (M!=I)
Shift 1 | Comps: 4 | Match!
Shift 2 | Comps: 1 | Mismatch (S!=I)
Shift 3 | Comps: 1 | Mismatch (S!=I)
Shift 4 | Comps: 4 | Match!
...

...Program finished with exit code 0
```

Question 4: Character Comparisons

main.c (C17)

```
#include <stdio.h>
#include <string.h>
int main() {
    char txt[] = "ABABABABAB", pat[] = "ABAB";
    int n = strlen(txt), m = strlen(pat);
    int comps = 0, matches = 0, mismatches = 0;
    for(int i = 0; i <= n - m; i++) {
        for(int j = 0; j < m; j++) {
            comps++;
            if(txt[i+j] == pat[j]) matches++;
            else { mismatches++; break; }
        }
    }
    printf("Total comparisons: %d\nTotal matches: %d\nTotal mismatches: %d\n",
        comps, matches, mismatches);
    return 0;
}
```

Output terminal

```
Text = "ABABABABAB"
Pattern = "ABAB"
Total comparisons: 18
Total matches: 14
Total mismatches: 4
```

...Program finished with exit code 0